

Uncovering the genetic architecture of parallel evolution

Maddie E. James^{1,2}  | Robin N. Allsopp¹  | Jeffrey S. Groh¹  | Avneet Kaur^{1,2}  |
Melanie J. Wilkinson^{1,2}  | Daniel Ortiz-Barrientos^{1,2} 

¹School of Biological Sciences, The University of Queensland, St Lucia, Queensland, Australia

²Australian Research Council Centre of Excellence for Plant Success in Nature and Agriculture, The University of Queensland, St Lucia, Queensland, Australia

Correspondence

Maddie E. James, School of Biological Sciences, The University of Queensland, St Lucia, QLD 4072, Australia.
Email: m.james4@uq.edu.au

Present address

Jeffrey S. Groh, Center for Population Biology and Department of Evolution and Ecology, University of California Davis, Davis, California, USA

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Abstract

Identifying the genetic architecture underlying adaptive traits is exceptionally challenging in natural populations. This is because associations between traits not only mask the targets of selection but also create correlated patterns of genomic divergence that hinder our ability to isolate causal genetic effects. Here, we examine the repeated evolution of components of the auxin pathway that have contributed to the replicated loss of gravitropism (i.e. the ability of a plant to bend in response to gravity) in multiple populations of the *Senecio laetus* species complex in Australia. We use a powerful approach which combines parallel population genomics with association mapping in a Multiparent Advanced Generation Inter-Cross (MAGIC) population to break down genetic and trait correlations to reveal how adaptive traits evolve during replicated evolution. We sequenced auxin and shoot gravitropism-related gene regions in 80 individuals from six natural populations (three parallel divergence events) and 133 individuals from a MAGIC population derived from two of the recently diverged natural populations. We show that artificial tail selection on gravitropism in the MAGIC population recreates patterns of parallel divergence in the auxin pathway in the natural populations. We reveal a set of 55 auxin gene regions that have evolved repeatedly during the evolution of the species, of which 50 are directly associated with gravitropism divergence in the MAGIC population. Our work creates a strong link between patterns of genomic divergence and trait variation contributing to replicated evolution by natural selection, paving the way to understand the origin and maintenance of adaptations in natural populations.

KEYWORDS

association mapping, auxin pathway, Multiparent Advanced Generation Inter-Cross (MAGIC) population, parallel evolution, recombinant population, shoot gravitropism

1 | INTRODUCTION

Natural selection explains the evolution of many traits in nature, yet identifying the genetic targets of selection remains a formidable

challenge (Barrett & Hoekstra, 2011; Hoban et al., 2016; Mackay et al., 2009; Ravinet et al., 2017; Rockman, 2012). Fortunately, natural systems where populations have repeatedly adapted to similar selective pressures (i.e. systems of parallel evolution (Schluter

Maddie E. James and Robin N. Allsopp equal first authors.

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& Nagel, 1995), also referred to as replicated evolution (James et al., 2023)) facilitate identifying genomic regions with shared signatures of selection across independent populations (Figure 1a). These repeated regions of high genetic differentiation across replicate populations are unlikely to have arisen by chance, suggesting that natural selection, rather than genetic drift, underlies their divergence (Losos, 2011; Orr, 2005; Schluter & Nagel, 1995). Despite the power of parallel evolution to identify putative targets of selection throughout the genome, it remains difficult to associate this genomic differentiation with patterns of phenotypic divergence (Barrett & Hoekstra, 2011; Hoban et al., 2016; Ravinet et al., 2017). This difficulty limits our ability to understand the genetics of adaptation in natural populations.

In systems of parallel evolution (e.g. Bohutínská et al., 2021; Butlin et al., 2014; Elmer et al., 2010; Jacobs et al., 2020; Stuart et al., 2017), patterns of shared genomic differentiation between replicate populations reflect the co-segregation of multiple traits and may reflect various sources of positive and negative selection (Stankowski et al., 2019). Divergence landscapes alone can therefore generally not be directly linked to particular traits of interest without multiple lines of evidence (Barrett & Hoekstra, 2011; Ravinet et al., 2017). Furthermore, neutral sites linked to alleles under selection will also exhibit elevated genomic divergence, making it difficult to infer which allele is under direct selection (Kaplan et al., 1989; Maynard Smith & Haigh, 1974). The few systems of parallel evolution that have successfully linked genotype to phenotype revealed

genetic architectures with one or a few genes of large effect (e.g. stickleback fish; Colosimo et al., 2005 and beach mice; Hoekstra et al., 2006). Identifying how parallel evolution proceeds for traits with more complex genetic architectures, where many genes underlie the adaptive phenotype, remains difficult for most natural systems.

In cases where multiple traits diverge in parallel, and/or trait divergence occurs between structured populations, association mapping in recombinant populations can provide a critical resource for identifying the alleles underlying variation in a trait of interest. For example, in Multiparent Advanced Generation Inter-Cross (MAGIC) populations (Mackay & Powell, 2007; Magwene et al., 2011; Mott et al., 2000), multiple generations of recombination allow genes and traits from parental populations to segregate (Broman, 2005). This recombination also breaks down linkage disequilibrium between causal and noncausal alleles due to population structure, thus enabling finer-scale mapping of the regions harbouring causal alleles (Barrett & Hoekstra, 2011). Loci that have an effect on any phenotype of interest can then be detected using a tail selection mapping approach, where shifts in allelic frequencies are compared between recombinant individuals in the opposite tails of a phenotypic distribution (Figure 1b), referred to as bulk segregant analysis (BSA; Ferris et al., 2015; Friedman & Willis, 2013; Gould et al., 2017; Magwene et al., 2011). A powerful approach, taken here, is to examine the loci identified through BSA association mapping for signatures of selection (e.g. F_{ST} outliers) in natural replicate populations.

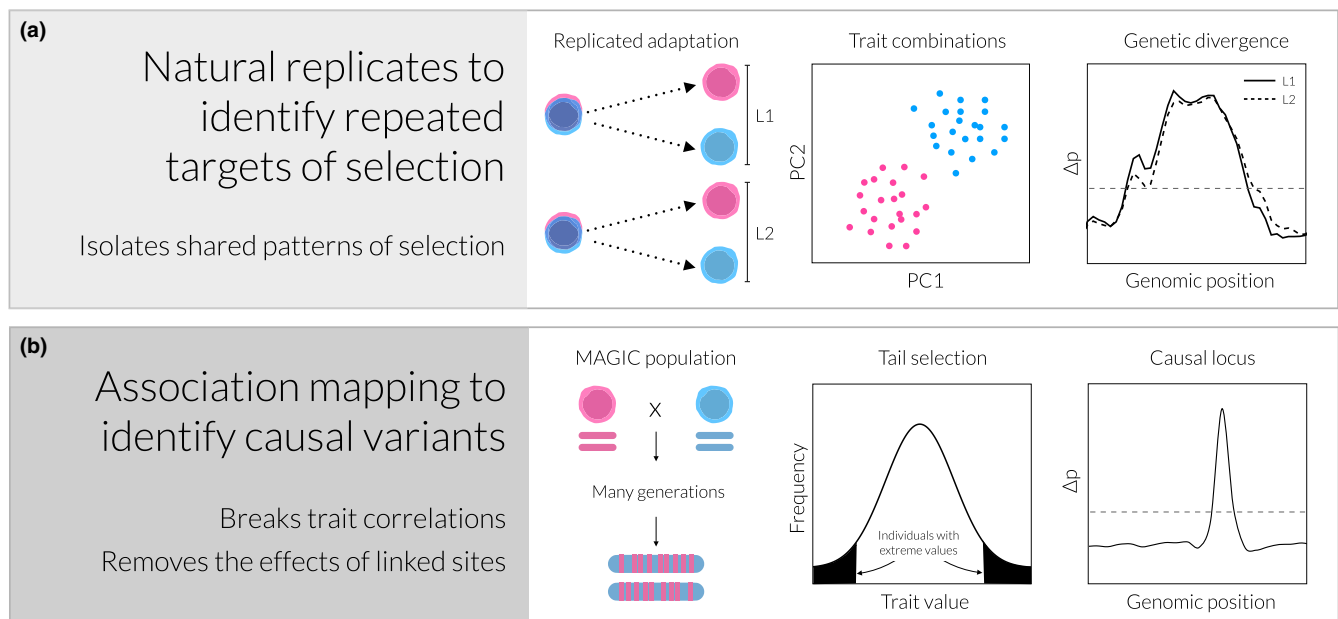


FIGURE 1 Uncovering the genetic architecture of parallel trait evolution. (a) Parallel population genomics. The independent evolution of divergent ecotypes (blue and pink circles) at two localities (L1 and L2). Ecotypes differ in many traits and cluster into distinct regions of multivariate trait space. Common regions of high differentiation (Δp) between ecotypes at each locality are unlikely to have arisen by chance; however, they represent the co-segregation of many traits and span a large genomic region. (b) Recombinant association mapping. Genomes of two populations are progressively broken up through many generations of crossing to create a Multiparent Advanced Generation Inter-Cross (MAGIC) population. Sequencing the extremes of a trait distribution (bulk segregant analysis) allows us to directly associate genomic variants with any trait of interest. As genetic linkage is broken, causal loci will exhibit a narrow peak of high differentiation (Δp). Dashed lines represent a genome-wide significance threshold.

The Australian wildflower species complex, *Senecio laetus*, consists of a variety of ecotypes adapted to contrasting environments. Two of these ecotypes, Dune and Headland, have evolved repeatedly and independently at multiple localities along the Australian coast (James, Arenas-Castro, et al., 2021). At each locality, there is a geographically adjacent Dune and Headland population, which we refer to as a Dune-Headland population pair (Figure 2a). Dune plants, which inhabit sheltered sandy dunes, exhibit an erect growth habit, while Headland plants occur on windswept rocky outcrops and have a prostrate growth habit (Figure 2a, Figure S2). These morphological differences between the ecotypes have a genetic basis (Radford et al., 2004; Walter, Aguirre, et al., 2018) and involve a multitude of traits (James, Wilkinson, et al., 2021; Walter, Aguirre, et al., 2018). The Dune and Headland ecotypes are obligate outcrossers and exhibit hybrid sterility that is weak in the F1 generation, yet strong in the F2 generation (Melo et al., 2014; Richards & Ortiz-Barrientos, 2016; Walter et al., 2016). Field selection experiments have revealed that local adaptation consistently creates strong extrinsic reproductive isolation between the two ecotypes at multiple localities (Melo et al., 2014; Richards et al., 2016; Richards & Ortiz-Barrientos, 2016; Walter et al., 2016; Walter, Wilkinson, et al., 2018; Wilkinson et al., 2021). Our ability to perform genetic experiments with *S. laetus* in both controlled and natural conditions (Melo et al., 2014; Richards & Ortiz-Barrientos, 2016; Walter et al., 2016) and our current understanding of the genetic basis of adaptation in the system (Walter

et al., 2020) provide an excellent opportunity to study the genetic basis of parallel adaptation.

Despite striking phenotypic similarities between populations of each ecotype in *S. laetus* (James, Wilkinson, et al., 2021), repeated selection on the same alleles or genes across localities is rare. Rather, different sets of genes that are enriched with similar biological functions appear to drive ecotypic divergence across localities (James, Wilkinson, et al., 2021; Roda et al., 2013). Notably, these sets of differentiated genes often have predicted functions in hormonal pathways, such as in the auxin pathway (Roda et al., 2013). Auxin is a phytohormone extensively involved in plant growth and development (Woodward & Bartel, 2005) and plays a fundamental role in plant movement, or tropism (Friml, 2003). Shoot gravitropism, the ability of plants to bend upwards in response to the downward pull of gravity, is an auxin-dependent trait and relies on establishing a local concentration gradient with high auxin levels along the lower side of the shoot (Muday, 2001; Figure 2b, Figure S1). Shoot gravitropism has diverged repeatedly in *S. laetus* (Wilkinson et al., 2021), where Dune plants respond strongly to gravity cues (gravitropic), and Headland plants respond weakly (agravitropic; Figure S2).

Shoot gravitropism shows a strong correlation with plant height across *S. laetus* ecotypes and directly responds to selection in the field (Wilkinson et al., 2021). As it is well established that shoot gravitropic responses in plants are governed by genes within the auxin pathway (Friml, 2003; Muday, 2001; Woodward & Bartel, 2005), we hypothesized that natural selection upon such auxin-related genes

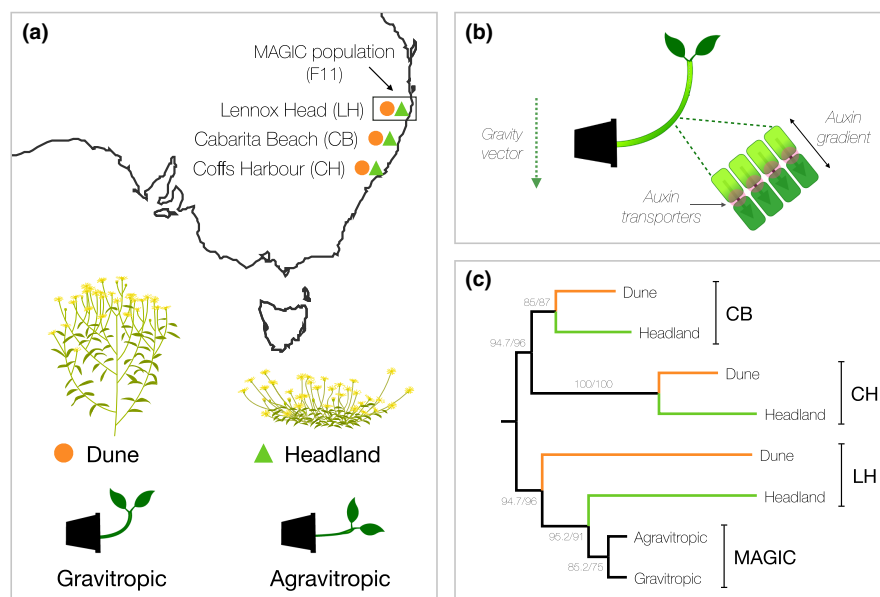


FIGURE 2 *Senecio laetus* system of parallel divergence. (a) Dune (orange circles) and Headland (green triangles) ecotypes at three replicate localities (Lennox Head—LH, Cabarita Beach—CB and Coffs Harbour—CH) along the coast of Australia. A Multiparent Advanced Generation Inter-Cross (MAGIC) population (F11 generation) was derived from ecotypes at Lennox Head. Dune plants are gravitropic (respond well to gravity cues) and Headland plants are agravitropic (respond poorly to gravity cues). (b) We focus on the role of the auxin hormone during the parallel evolution of shoot gravitropism divergence (also see Figure S1). Re-orientation of a plant by 90° results in auxin accumulation on the lower side of the hypocotyl. High concentrations of auxin cause cells to elongate, leading to curvature of the hypocotyl and a gravitropic response. (c) Maximum likelihood phylogeny of Dune and Headland populations and the MAGIC population of unlinked SNPs from the 538 auxin genes, implemented in IQ-TREE. Numbers on each node represent the SH-aLRT support (%), followed by the ultrafast bootstrap support (%).

could have contributed to morphological differentiation between ecotypes. Consistent with this knowledge of the molecular components underlying gravitropism, Wilkinson et al. (2021) recently undertook genetic work in a recombinant MAGIC population constructed from the Dune and Headland ecotypes at one locality and identified two large effect loci that were associated with gravitropism divergence and located in regions of the genome containing gene homologues with functions related to the auxin pathway. However, this work used reduced representation sequencing and likely did not sequence all genes in the auxin pathway. It thus remains unclear what other auxin-related genes might have contributed to shoot gravitropism divergence in *Senecio*, and what core components of the auxin pathway could have been repeatedly selected for across replicate Dune-Headland population pairs. Here, we tackle these issues by combining genetic signatures of parallel evolution with association mapping in the same recombinant MAGIC population as Wilkinson et al. (2021) to directly evaluate whether variation in the auxin and shoot gravitropism pathways contributes to the repeated adaptive divergence in *S. lautus*. Our work sheds new light on how natural selection repeatedly acts on genetic components of a signalling pathway essential to the evolution of a complex trait during adaptive divergence.

2 | MATERIALS AND METHODS

2.1 | *Senecio lautus* samples

Leaf samples from six natural populations of *S. lautus* (adjacent Dune-Headland population pairs from three localities: Lennox Head, Cabarita Beach and Coffs Harbour, $N_{\text{total}}=80$; Table S1) were collected as part of a separate investigation conducted by James, Arenas-Castro, et al. (2021), see Supplementary Methods for details. The Multiparent Advanced Generation Inter-Cross (MAGIC) population (F11 generation, $N=1200$ diploids) used in this study was produced and shoot gravitropism measurements taken as part of a previous investigation by Wilkinson et al. (2021), see Supplementary Methods for details. Briefly, this population was derived from 23 Dune and 22 Headland individuals from Lennox Head, randomly mated with equal contribution for eight generations before undergoing a field selection experiment for an additional three generations. Gravitropism was measured on the resulting F11 population by rotating seedlings 90° and measuring the change in angle after 12 h. For this study, individuals from the tails of the F11 gravitropism distribution were selected for DNA extraction—70 'gravitropic' individuals where the change in angle upon rotation was $>56.6^\circ$, and 70 'agravitropic' individuals where the change in angle upon rotation was $<21.4^\circ$.

2.2 | Auxin gene regions

As our focus in this study is understanding the evolution of an adaptive trait, shoot gravitropism, we focussed on genes known to directly

control the regulation of shoot bending in plants (Friml, 2003; Muday, 2001; Woodward & Bartel, 2005). To do this, we compiled a comprehensive set of genes involved in the auxin pathway and shoot gravitropism through a rigorous review in the model species *Arabidopsis thaliana*, where extensive functional experiments have been conducted. Guided by the book *Hormone Metabolism & Signaling in Plants* (Li et al., 2017), review papers (Leyser, 2006, 2018; Ljung, 2013; Ludwig-Müller, 2011; Petrásek & Friml, 2009; Strohm et al., 2012; Zhao, 2014), and The Arabidopsis Information Resource (TAIR) database, we curated a set of well-characterized auxin and shoot gravitropism genes. For each gene, we used primary literature to critically evaluate relevance for inclusion and summarized the type, strength and source of evidence. Through this method, we curated a robust set of 354 genes involved in the auxin pathway and/or shoot gravitropism in *A. thaliana* (see Table S2). This functional validation gives us a high degree of confidence that the genomic regions we are studying are directly associated with our adaptive trait of interest. However, it is important to note that our approach does not consider auxin-related genes that are not well characterized, as well as any novel *Senecio* auxin and shoot gravitropism genes. We do not claim to reveal the entire genetic architecture of shoot gravitropism in this system. Instead, our work aims to uncover key molecular components from a pathway known to influence gravitropism in plants. By examining this auxin-regulated trait, we aim to identify the specific components of the auxin and shoot gravitropism pathways that contribute to adaptation in *Senecio*.

We grouped each gene into functional categories based on its primary role in the auxin pathway or shoot gravitropism (Figure S1). We classified each gene into a specific functional category to gain a deeper understanding of the physical processes through which auxin triggers a gravitropic response in *Senecio*. This classification also helps us determine whether the overall mechanisms of shoot gravitropic responses are consistent across the naturally occurring instances of evolution. It is important to note that these functional categories are a simplification of a complex and interrelated pathway. In this study: 'biosynthesis' is the production of auxin (47 genes); 'conjugation' is the alteration of auxin functional properties via structural modification and/or coupling/uncoupling of substrates (35 genes); 'transport' is the active movement of auxin within and between cells (29 genes); 'localization' is the dynamic control of auxin transporter position and activity (46 genes); 'signalling' is the auxin-dependent regulation of gene expression (162 genes); and gravitropism is the nonauxin components of shoot gravity perception/response (35 genes).

Using this gene set from *A. thaliana*, we identified orthologous gene regions in the *S. lautus* PacBio genome draft (James, Arenas-Castro, et al., 2021). Given the divergence time between these species is >100 mya (Moore et al., 2010), we used a species phylogenetically closer to *S. lautus* to improve confidence in orthologous matches. For this purpose, we chose a member of the same family (Asteraceae), *Helianthus annuus*, as it has a well-annotated genome and diverged from *S. lautus* <45 mya (Panero & Crozier, 2016). First, we found the closest match in *H. annuus* for each *A. thaliana*

protein from our original gene set using a BLASTp search, ranking by *e*-value. We then searched each *H. annuus* protein against the nucleotides of the *S. laetus* reference genome, using tBLASTn. To account for differences in gene family size between species, in cases where there were more high-quality matches in *S. laetus* than members of a gene family in the *A. thaliana* set, additional matches were listed as duplicates of the gene they most closely resembled. Genes with poor matches (<50% identity and/or <50% coverage) between species at any step were excluded, leaving 596 putative *S. laetus* auxin and shoot gravitropism gene regions in the final annotation. We refer to these regions hereafter as 'auxin genes'. We note that due to this annotation approach (which was undertaken because we lack an annotated reference genome for *S. laetus*), these gene regions may also contain pseudogenic artefacts, multiple genes from the same family located close together in the genome and/or additional nonauxin or shoot gravitropism genes overlapping the same region.

2.3 | DNA extraction and sequencing

We aimed to sequence 80 genomes from six natural populations, and 70 gravitropic and 70 agravitropic individuals from the tails of the gravitropism distribution of the MAGIC population. For the natural populations, total genomic DNA was extracted from leaf tissue using a modified cetrimonium bromide (CTAB) protocol (Clarke, 2009), see Supplementary Methods. For the MAGIC population, total genomic DNA was extracted from leaf tissue using the Qiagen DNeasy Plant Pro kit as per the manufacturer's instructions for samples high in polyphenolic compounds. We quantified DNA concentration with the Invitrogen Quant-iT PicoGreen dsDNA Assay Kit to ensure that samples contained >500 ng of DNA at a concentration of >12.5 ng/μL. The BioTek Take3 Micro-Volume Plate was used to ensure DNA samples were pure (260/280 between 1.8 and 2.0 and 260/230 between 2.0 and 2.2). Samples that did not meet these standards were excluded. For the natural populations, all 80 samples were retained; for the MAGIC population, 70 gravitropic and 63 agravitropic samples were retained. DNA samples were arranged randomly with respect to population (for natural populations) or gravitropism tail (for the MAGIC population) on the 96-well plates submitted for individual-based sequencing. Library preparation and sequencing was conducted by Beijing Genomics Institute (BGI) using the DNBseq platform, 100 bp paired-end reads, at ~5X depth.

We note it is common for studies undertaking bulk segregant analysis to perform pooled sequencing of the tails of a phenotypic distribution (e.g. Ferris et al., 2015; Friedman & Willis, 2013; Gould et al., 2017). In this work, we undertook individual-based sequencing to gain more confidence in calling low-frequency variants and bypass issues associated with optimal amount of sequencing coverage per bulk segregant pool (see Magwene et al., 2011).

2.4 | Bioinformatics

Scripts and detailed explanations of all bioinformatics and analyses are available on GitHub (<https://github.com/OB-lab/James-et-al.-2023-MolecularEcology>). For both the natural populations and MAGIC population datasets, raw sequencing data was cleaned by BGI to remove barcode sequences, DNBseq adaptors, low-quality reads (>50% of quality scores <10) and reads containing >10% unidentified bases. This left an average of 94,350,329 clean reads (SD=3,073,792) per individual for the natural populations and 87,188,853 clean reads per individual (SD=10,000,145) for the MAGIC population. Reads were aligned to the *S. laetus* reference genome (James, Arenas-Castro, et al., 2021) and read groups added with the short-read BWA-MEM v0.7.13 (Li & Durbin, 2009) algorithm under default parameters. For natural populations, an average of 97.7% (SD=1.08) of reads mapped to the reference genome, 90.6% (SD=2.26) with their paired read. For the MAGIC population, an average of 98.6% (SD=0.6) of reads mapped to the reference genome, 92.6% (SD=2.3) with their paired read. Binary alignment map (BAM) files were sorted with Samtools v1.3 (Li et al., 2009). For the MAGIC population, sorted BAM files were cleaned using PicardTools v2.22 (Broad Institute, 2019) CleanSam to softclip reads extending beyond the reference genome and set unmapped quality scores to 0. For both datasets, PCR duplicates were marked for removal: for natural populations, this was done with Samblaster v0.1.24 (Faust & Hall, 2014) with default parameters; in the MAGIC population, this was done with PicardTools MarkDuplicates with default parameters. Regions around indels were realigned using GenomeAnalysisToolKit v3.8 (Van der Auwera & O'Connor, 2020) IndelRealigner. Final BAM files were indexed using Samtools and validated using PicardTools ValidateSamFile.

Allele frequencies were calculated for all sites (both variant and invariant) across the 596 auxin genes per natural population and per tail of the MAGIC population in ANGSD v0.932 (Korneliussen et al., 2014). The GATK genotype likelihood model was used, and sites were filtered out if the per-base quality score was <20 and the mapping quality was <30. Allele frequency files for each population were combined using custom scripts in R v4.1.0 (R Core Team, 2017). We retained sites if they were sampled in at least three individuals in each natural population and at least five individuals in each tail of the MAGIC population, in which 538 auxin genes were retained. To filter for a MAF>0.05 while retaining invariant sites, we used custom R scripts to replace all allele frequencies that are less than 0.05 with 0, which was done per population or tail. All figures were made in JMP v16.2 unless otherwise specified.

2.5 | Genetic clustering of populations

To uncover the genetic relationships between populations, we performed two separate analyses: phylogeny construction and

principal components analysis (PCA). We generated a maximum likelihood phylogeny in *IQ-TREE* v1.6.12 (Nguyen et al., 2015) using the polymorphisms-aware phylogenetic model (Schrempf et al., 2016). We used sites that were variable across all populations and the MAGIC population with a $MAF > 0.05$. We used custom R scripts to thin SNPs by retaining one SNP per auxin gene. We first used ModelFinder (Kalyaanamoorthy et al., 2017) to determine the best-fit substitution model for the data (TVMe + FQ + P + N9). Default parameters were used for tree construction. To assess convergence, we undertook 10 separate runs of *IQ-TREE* and examined tree topology (which remained unchanged across the 10 independent runs). We also ensured that the log-likelihood values were stable at the end of each run. Branch support was performed using 10,000 replicates of UFboot (Hoang et al., 2018) and 10,000 replicates of SH-aLRT (Guindon et al., 2010). In addition, we repeated the above process five times using a different set of unlinked SNPs each time. Results remained consistent across each of the five unlinked datasets (data not shown). Data were visualized in FigTree v1.4.4 (Rambaut, 2007).

We further explored the genetic clustering of populations using PCAngsd v1.10 (Meisner & Albrechtsen, 2018), which estimates the covariance matrix based on the genotype likelihoods of samples. The GATK genotype likelihood model was used, and sites were filtered out if the per-base quality score was < 20 and the mapping quality was < 30 . The p -value for detecting a SNP was set to 1×10^{-6} . As above, we thinned SNPs by retaining one unlinked SNP per auxin gene and repeated this process five times using a different set of unlinked SNPs each time. Results remained consistent across each of the five unlinked datasets (data not shown). Data were plotted in R.

2.6 | Parallel evolution in the auxin pathway

We used a combination of three approaches to detect outlier SNPs across the auxin and shoot gravitropism gene regions between the natural Dune and Headland populations per locality: PCAngsd, BayeScan v2.1 (Foll & Gaggiotti, 2008) and the top 1% of SNPs with the highest change in allele frequencies between the Dune and Headland. PCAngsd estimates the covariance matrix based on the genotype likelihoods of samples. We calculated genotype likelihoods per locality using the GATK genotype likelihood, and sites were filtered out if the per-base quality score was < 20 , and the mapping quality was < 30 . We only retained variant sites, those with a p -value threshold of 1×10^{-6} . Using PCAngsd, we calculated a selection statistic for each SNP as undertaken in FastPCA (Galinsky et al., 2016). Selection statistics were converted to p -values using the pchisq function in R. For each locality, we selected the single principal component (PC) that best separated the ecotypes and considered a SNP as highly differentiated if it had a p -value < 0.002 on the selected PC axis (see <https://github.com/OB-lab/James-et-al.-2023-MolecularEcology> for further details).

BayeScan implements a reversible-jump MCMC algorithm to estimate posterior odds comparing a model with and without

selection for each SNP. We applied BayeScan per locality on sites with a $MAF > 0.05$. We used default parameters with a prior odds of ten, meaning the neutral model is ten times more likely than the model with selection (as recommended for datasets of our size; Foll & Gaggiotti, 2008). We categorized SNPs as highly differentiated if they contained a Q -value (the False Discovery Rate analogue of the p -value) < 0.05 .

For our third approach, we calculated the absolute allele frequency difference ($|\Delta p|$) for each allele between the Dune and Headland ecotypes at each locality and retained the top 1%. Again, we only considered sites with a $MAF > 0.05$ per locality. We classified SNPs as highly differentiated if they had an allele frequency difference in the top 1% quantile at each locality (corresponding to $|\Delta p| \geq 0.63$ for Lennox Head, $|\Delta p| \geq 0.53$ for Cabarita Beach and $|\Delta p| \geq 0.52$ for Coffs Harbour).

For each locality, we considered a SNP as an outlier if it was detected as highly differentiated in at least one of the three above approaches. For the stringent dataset, SNPs were outliers if they were highly differentiated in at least two of the three approaches. For each locality, we considered a gene as an outlier if it contained at least one outlier SNP. We compared the number of common outlier SNPs and genes across the three localities and classified a SNP or gene as 'parallel' if it was an outlier in all three localities; we refer to these as 'parallel SNPs' and 'parallel genes', respectively. We also calculated whether the number of shared outlier SNPs and genes between the localities was greater than expected by chance using a hypergeometric distribution function, *phyper*, in R, which calculates the probability of obtaining overlap between groups of predefined sizes (see <https://github.com/OB-lab/James-et-al.-2023-MolecularEcology> for further details).

To identify molecular signatures of selection for the 55 parallel genes as well as the three parallel genes from the stringent dataset, we used ANGSD to calculate Tajima's D within each of the natural populations for each gene. We incorporated invariant sites to provide accurate estimates of nucleotide diversity (Korunes & Samuk, 2021).

2.7 | Linking phenotype to genotype with the MAGIC population

To uncover which of the auxin and shoot gravitropism genes contribute to gravitropism divergence, we analysed patterns of genetic differentiation between individuals from the agravitropic and gravitropic tails in the MAGIC population. We omitted sites that were sampled in fewer than five individuals in either the gravitropic or agravitropic tail. To detect outlier SNPs between the tails, we employed a combination of two methods: Fisher's exact test and the identification of the top 1% of SNPs with the greatest change in allele frequencies between the tails. While we also used PCAngsd and BayeScan to detect outliers in the natural populations, these methods were not suitable for a highly recombined population. Specifically, PCAngsd did not reveal any principal

components that separated the gravitropic and agravitropic tails. Additionally, BayeScan, which is designed for natural populations that exhibit greater genetic divergence, was unable to detect the subtle shifts in allele frequencies between the two tails of the MAGIC population.

Fisher's exact test (FET) is a statistical method used to determine the exact probability of an association between the observed and expected frequencies for two categorical variables (Fisher, 1922). Given its suitability for binary SNP data and its robust performance with small sample sizes and uneven data distribution (Choi et al., 2015; Fisher, 1946; Kim, 2017), FET is an ideal choice for our analysis. We used FET to investigate whether there are significant differences between the agravitropic and gravitropic tails at each SNP. To conduct the FET, we first constructed a 2×2 contingency table of the allele counts for the upper and lower tails for each SNP. We then performed the FET in R. To correct for multiple testing, we applied the Benjamini–Hochberg procedure, which effectively controls the false discovery rate (Benjamini & Hochberg, 1995). We classified SNPs as highly differentiated if they had an adjusted p -value of less than 0.05.

It is worth noting that bulk segregant analysis studies typically employ the G-test (Magwene et al., 2011). This is a likelihood ratio test that, similar to FET, determines whether there is a significant difference between observed and expected frequencies across two or more categories. However, the G-test is not recommended for sample sizes larger than ten in each cell and does not perform as well as FET when data are unevenly distributed (Magwene et al., 2011). Given that the site frequency spectrum of the MAGIC population is heavily enriched with loci having very similar frequencies between the tails, many loci present uneven distributions of allele counts, and cell counts are sometimes less than ten, we opted to use FET (as described above). Nevertheless, we also examined the relationship between the G-test and FET. We used the G-test function in DescTools v0.99.49 (Signorell et al., 2017) in R to calculate the G-statistic and associated p -value for the 2×2 contingency tables of the allele counts for the upper and lower tails for each SNP. We then used R to plot the relationship between the G-statistic and FET p -values.

For the top 1%, we calculated the absolute allele frequency difference ($|\Delta p|$) between the gravitropic and agravitropic tails, and as above, we considered SNPs as highly differentiated if they had an allele frequency difference in the top 1% quantile (corresponding to $|\Delta p| \geq 0.18$). These allelic frequencies mirror an independent reduced representation dataset on the tails of the gravitropism distribution where the allelic difference in the 99.8% quantile was $0.15 < p < 0.22$ (Wilkinson et al., 2021). For the MAGIC population, we considered a SNP as an outlier if it was detected as highly differentiated in at least one of the two above approaches (FET and top 1%); we refer to these SNPs as 'architecture SNPs'. For the stringent dataset, SNPs were outliers if they were highly differentiated in both approaches. For the MAGIC population, we considered a gene as an outlier if it contained at least one outlier SNP; we refer to these genes as 'architecture genes'.

2.8 | Integrating parallel population genomics with the MAGIC population

We then combined insights from the natural populations with the MAGIC population. Specifically, we asked which of the parallel SNPs and parallel genes in the natural populations were also outliers in the MAGIC population. We refer to these as 'parallel architecture SNPs' or 'parallel architecture genes', respectively. As above, we used a hypergeometric distribution function, *phyper*, in R to calculate whether the number of parallel architecture SNPs and parallel architecture genes was greater than expected by chance.

We note that the various approaches used in our work disregard allelic effect sizes, which would provide a more comprehensive view of trait architecture from a known molecular pathway. It is likely that we sampled many alleles of large effects given that we chose a high threshold to define outlier SNPs for the natural and MAGIC populations, and tail selection in the MAGIC population biases isolation of alleles with moderate-to-large effect sizes.

2.9 | Functional category enrichment

For each of the above sets of outlier genes (Lennox Head, Cabarita Beach, Coffs Harbour, parallel genes, architecture genes and parallel architecture genes), we calculated whether any functional categories are overrepresented in relation to our auxin and shoot gravitropism gene set. To calculate the fold change, for each set we calculated the proportion of outlier genes in each functional category (biosynthesis, conjugation, transport, localization, signalling or gravitropism) and divided it by the proportion of genes for that functional category that are in our auxin and shoot gravitropism gene set. We also undertook this for the stringent dataset. We note that the functional categories of transport and localization are inherently related: we maintained separate categories to distinguish between the direct transporter machinery (transport) and other cellular components that influence activity and orientation of transporters (localization).

3 | RESULTS

3.1 | Auxin and shoot gravitropism gene regions

We curated a set of well-characterized auxin and shoot gravitropism genes that are directly involved in shoot gravitropism through a rigorous review of published literature in the model species *Arabidopsis thaliana* (see Table S2 for evidence for each gene). These genes were grouped into functional categories (biosynthesis, conjugation, transport, localization, signalling or gravitropism) based on their primary role in the auxin pathway or shoot gravitropism (Table S2, Figure S1). We identified their orthologous gene regions in the *S. lautus* genome (James, Arenas-Castro, et al., 2021) and obtained a

set of 596 candidate gene regions, which we refer to as 'auxin genes'. We extracted these auxin genes from the genomes of 80 individuals from three replicate Dune-Headland natural population pairs as well as 133 individuals from the tails of the gravitropism distribution of the MAGIC population (Figure 2a; Table S1). Of these 596 auxin genes, 538 were successfully sequenced across all natural populations ($N_{pop} \geq 3$) and the MAGIC population ($N_{tail} \geq 5$). These 538 genes contained 250,166 variable sites (MAF > 0.05). Phylogenetic analysis recapitulates previously estimated phylogenetic relationships within *S. laetus* from genome-wide markers, further supporting multiple origins of the two ecotypes and suggesting that variation in these genes does not reflect recent introgression among populations of the same ecotype (Figure 2c, Figure S3; James, Arenas-Castro, et al., 2021).

3.2 | Parallel evolution in the auxin pathway

We assessed the extent of divergence in the 538 auxin genes from the three replicate Dune-Headland natural population pairs at different levels of biological organization: the SNP, gene and biological function. For these 538 genes, we used a combination of approaches (PCAngsd, BayeScan and the top 1% of SNPs with the highest change in allele frequencies between the Dune and Headland) to

detect highly differentiated (or 'outlier') SNPs per locality (see Methods; Figure S4). In addition, we created a 'stringent' dataset to examine the robustness of results (see Methods for further details). In the auxin and shoot gravitropism gene set, we detected 1684 outlier SNPs for Lennox Head, 1061 for Cabarita Beach and 1085 for Coffs Harbour (Figure 3a). The number of outlier auxin genes (i.e. genes containing at least one outlier SNP), was quite similar for each locality: 179 outlier genes for Lennox Head, 145 for Cabarita Beach and 137 for Coffs Harbour (Figure 3a). For each locality, these outlier genes were enriched in the auxin transport functional category (i.e. the active movement of auxin within and between cells) relative to other auxin-related functions (Figure 3b). This consistent enrichment suggests that the outlier genes per locality perform similar biological functions and that natural selection has repeatedly acted upon genes that control the redistribution, or transport, of auxin within the cell to elicit a gravitropic response. These results held true for the stringent dataset, where the numbers of outliers were similar across localities, and outlier genes were also enriched in the auxin transport functional category (Figure S5). Because outliers of genetic differentiation can also reflect genetic drift (Buerkle et al., 2011; Ravinet et al., 2017), we next focussed our attention on SNPs and genes that showed repeated differentiation across all three localities and are therefore most likely to reflect divergence via natural selection (Orr, 2005).

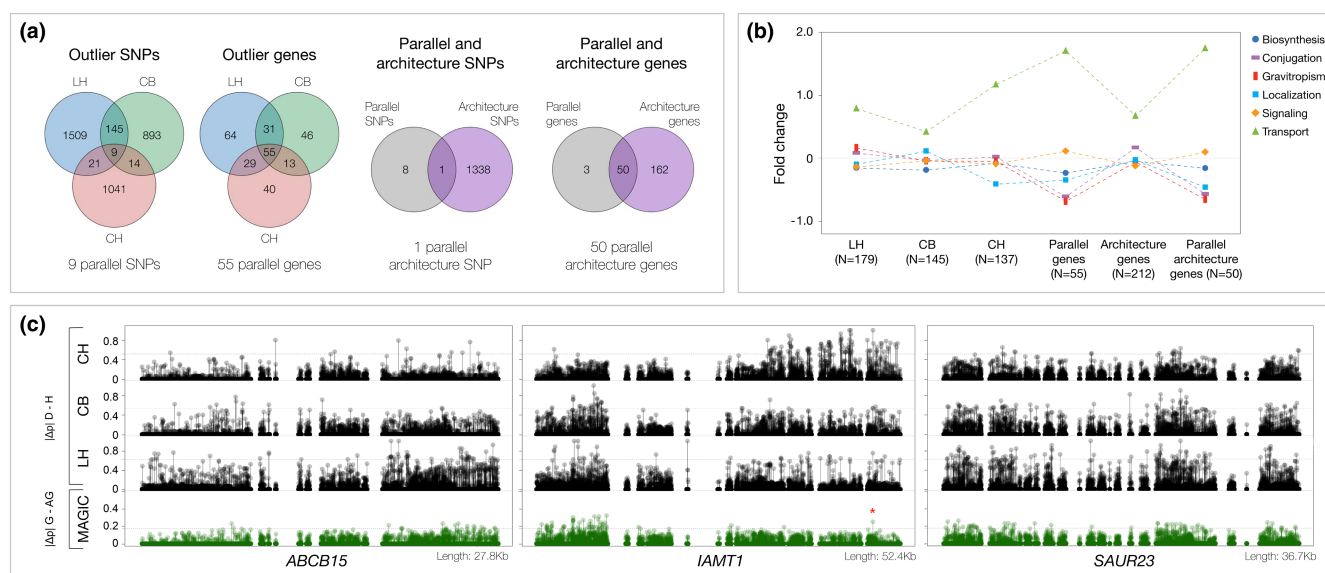


FIGURE 3 Patterns of genomic divergence across natural populations and the MAGIC population. (a) Two left Venn diagrams: number of outlier SNPs and outlier genes across replicate localities. Two right Venn diagrams: number of parallel and architecture SNPs, and parallel and architecture genes. Parallel SNPs/genes are outliers in all three natural localities. Architecture SNPs/genes are outliers in the MAGIC population. (b) Functional categorization of outlier genes. Fold change represents whether outlier genes are disproportionately represented in a particular functional category (proportion of outlier genes divided by the proportion of genes in the auxin and shoot gravitropism gene set for each functional category). 'Parallel genes' are outliers in all three natural localities, 'architecture genes' are outliers in the MAGIC population, and 'parallel architecture genes' are outliers in both sets. N represents the number of outlier genes. (c) Absolute allele frequency differences (Δp) in three natural localities (top three panels; Dune (D) and Headland (H); Lennox Head—LH, Cabarita Beach—CB and Coffs Harbour—CH) and the F11 MAGIC population (bottom panel; gravitropic (G) and agravitropic (AG)) for the top three candidate parallel architecture genes. Dashed grey lines represent the top 1% threshold of SNPs with the highest change in allele frequencies between the Dune and Headland (LH=0.63, CB=0.53, CH=0.52, MAGIC=0.18). The red asterisk represents the parallel architecture SNP. These three genes were outliers in the stringent dataset.

We discovered nine 'parallel SNPs' (those that were outliers in all three localities; Figure 3a), which were in four genes (*SGR9*, *IAMT1*, *SPK1*, *SAUR64*; see Table S3). We discovered a set of 55 'parallel genes' that contained outliers of genetic differentiation (i.e. at least one highly differentiated SNP) across all three localities (Figure 3a, Table S3). Consistent with a role for natural selection in driving their evolution in *S. lautus*, most of these parallel genes carry an excess of rare variants, which is particularly pronounced in the Lennox Head population (as seen by negative Tajima's *D* values; Figure S6A). Mirroring the results of the outlier genes for each locality, this set of parallel genes is also enriched in functions related to auxin transport (Figure 3b). For the stringent dataset, we detected no parallel SNPs and three parallel genes (Figure S5A). Again, results were consistent across datasets, where the parallel genes in the stringent dataset were enriched in functions related to auxin transport (Figure S5B) and showed signatures of selection, mostly in Lennox Head (Figure S6B). The parallel SNPs and genes from both datasets are highly unlikely to have arisen by chance across natural populations (Table S4), implying they have likely been repeated targets of natural selection on the auxin pathway in the system.

3.3 | Linking phenotype to genotype with the MAGIC population

To directly link genetic variation in *S. lautus* auxin genes to divergence in shoot gravitropism, we used a previously established Multiparent Advanced Generation Inter-Cross (MAGIC) population (F11 generation, *N* = 1200 diploids) derived from the Dune and Headland ecotypes at the Lennox Head locality (Figure 2a; Table S1; Wilkinson et al., 2021). A gravitropism assay was previously performed on this population—seedlings were rotated 90° relative to the gravity vector, and shoot gravitropism was measured by the change in angle of the shoot after 12 h (Wilkinson et al., 2021). We assessed the extent of genetic differentiation across the 538 auxin genes between the tails of the gravitropic distribution of the MAGIC population: 70 'gravitropic' individuals that responded strongly to gravity (change in angle >56.6°) and 63 'agravitropic' individuals that responded weakly or not at all to gravity (change in angle <21.4°).

In accordance with theoretical expectations for a MAGIC population (Magwene et al., 2011), the mean allele frequency difference between gravitropic and agravitropic tails was close to zero ($|\Delta p| = 0.002$, *SD* = 0.014). To detect outlier SNPs between the gravitropic and agravitropic tails of the MAGIC population, we used a combination of two approaches (FET and the top 1% of SNPs with the highest change in allele frequencies between the agravitropic and gravitropic tails) to detect highly differentiated (or 'outlier') SNPs for the MAGIC population (see Methods; Figure S7). (Although for the natural populations we used a combination of three approaches to detect outlier SNPs, two of these approaches were not appropriate for a highly recombined population, see Methods for further details). In addition, we created a 'stringent' dataset to examine the robustness of results (see Methods for further details). Furthermore,

we explored the relationship between the *p*-values of FET and the *p*-values of the G-statistic and found strong similarities between the two tests (Figure S8), highlighting the robustness of our results for the chosen statistical method. In fact, the majority of *p*-values from the G-statistic are smaller than *p*-values from FET, implying our conclusions about parallelism in *Senecio* are conservative.

We detected 1339 outlier SNPs in the MAGIC population that were differentiated in at least one of the two approaches, we refer to these loci as 'architecture SNPs'. We found 212 genes contained at least one of the highly differentiated SNPs. We refer to these loci as 'architecture genes': auxin and shoot gravitropism genes that explain differences between the individuals from the two tails of the gravitropic distribution of the MAGIC population. Mirroring our results from the natural populations and parallel genes, architecture genes were also enriched in the auxin transport functional category (Figure 3b). Thus, our artificial selection on the extremes of gravitropism captured the same functional variation underlying repeated differentiation in the auxin pathway caused by local adaptation to contrasting environments. For the stringent dataset, we detected 163 architecture SNPs and 47 architecture genes (Figure S5A). Results were consistent across datasets, where the architecture genes in the stringent dataset were also enriched in functions related to auxin transport (Figure S5B). To reveal which, if any, of the genes underlying gravitropism were repeatedly recruited during the divergence of *S. lautus* ecotypes, we integrate the genetic architecture of shoot gravitropism (from the MAGIC population) with patterns of genomic parallelism in the auxin pathway in *S. lautus* natural populations.

3.4 | Integrating parallel population genomics with the MAGIC population

Of the nine parallel SNPs that were highly differentiated across all three natural population pairs, one SNP was also highly differentiated between the tails of the MAGIC population (Figure 3a, asterisk in *IAMT1* Figure 3c, Table S3). We refer to this SNP as a 'parallel architecture SNP'. Remarkably, of the 55 parallel genes detected in the natural populations, 50 were also outliers in the MAGIC population, which we refer to as 'parallel architecture genes' (Figure 3a, Figure S9). These 50 parallel architecture genes were also enriched in the auxin transport functional category (Figure 3b). Taken together with the above results, this reveals that the transport functional category is consistently enriched across all sets of outlier genes (Lennox Head, Cabarita Beach, Coffs Harbour, parallel genes, architecture genes and parallel architecture genes). Although the two functional categories of transport and localization are inherently related, when combined they still show enrichment across all sets of outlier genes (data not shown). This demonstrates that our functional enrichment results are robust and suggests that the movement of auxin is a crucial factor in the parallel genetic architecture of gravitropism divergence. In the stringent dataset, all three parallel genes in the natural populations were also outliers in the MAGIC population (Figure S5). These three genes are our top candidates contributing to the parallel

evolution of shoot gravitropism in *S. laetus* (Figure 3c). The parallel architecture SNP and genes are highly unlikely to have arisen by chance across populations (Table S5).

We explored the within-gene patterns of genetic variation in the MAGIC population across the 50 parallel architecture genes. For these 50 genes, average allele frequency differences in the MAGIC population were lower than in the natural populations (Figure 4a). The higher allele frequency differences in the natural populations might suggest that many differentiated SNPs have hitchhiked to high frequencies along with the allele/s under selection in each locality. Furthermore, for each of the 50 parallel architecture genes we found there were often fewer outlier SNPs in the MAGIC population than in the natural populations (Figure 4b, Figure S10). For some of these parallel architecture genes, genetic differentiation in the MAGIC population was restricted to small regions of the auxin genes, sometimes single SNPs, suggesting a disassociation between noncausative and causative loci in the MAGIC population (Figure S9). For example, in *PIN4*, *SAUR28* and *SGR9* (Figures S9 and S10) we can identify single SNPs associated with gravitropism despite quite high divergence across the entire genic region in most natural populations. This is not the case for other genes such as *SAUR23* (Figure 3c), where patterns of genomic divergence in the MAGIC population recapitulated the same patterns found in the natural populations, likely due to scarcity of recombination events within these genes during the creation of the MAGIC population. Overall, integrating replicates of adaptation with genetic mapping in a MAGIC population is helping

us to overcome the effects of linked selection and reveal the partial genetic architecture of a trait.

4 | DISCUSSION

Disentangling the genetic architecture of adaptive traits in natural populations is challenging because evolution is often complex at both the genotypic and phenotypic levels. Although quantitative trait loci (QTL) mapping between two different parents has been successful in associating genomic regions with traits of interest, this approach is often underpowered to pinpoint genetic variants to a high resolution (Mackay et al., 2009). This makes it difficult to identify the genetic architecture of a trait in natural populations. Of the few QTL mapping studies that identified causal loci of adaptation, genetic architectures were often simple and consisted of few genes with alleles of large effect (e.g. Chan et al., 2010; Colosimo et al., 2005; Ferris et al., 2017; Hoekstra, 2006). In model organisms like *Arabidopsis* (Ågren et al., 2017; Fournier-Level et al., 2011; Stinchcombe et al., 2004) and *Drosophila* (Huang et al., 2014; Mackay et al., 2012), loci responsible for adaptation have been identified, but these systems often rely on genetic resources that are not available in other organisms (but see the work in sunflowers (Todesco et al., 2020; Whitney et al., 2015), sticklebacks (Colosimo et al., 2005; Schluter et al., 2021) and *Mimulus* (Lowry & Willis, 2010; Wu et al., 2008)). Fortunately, the natural replication in systems of

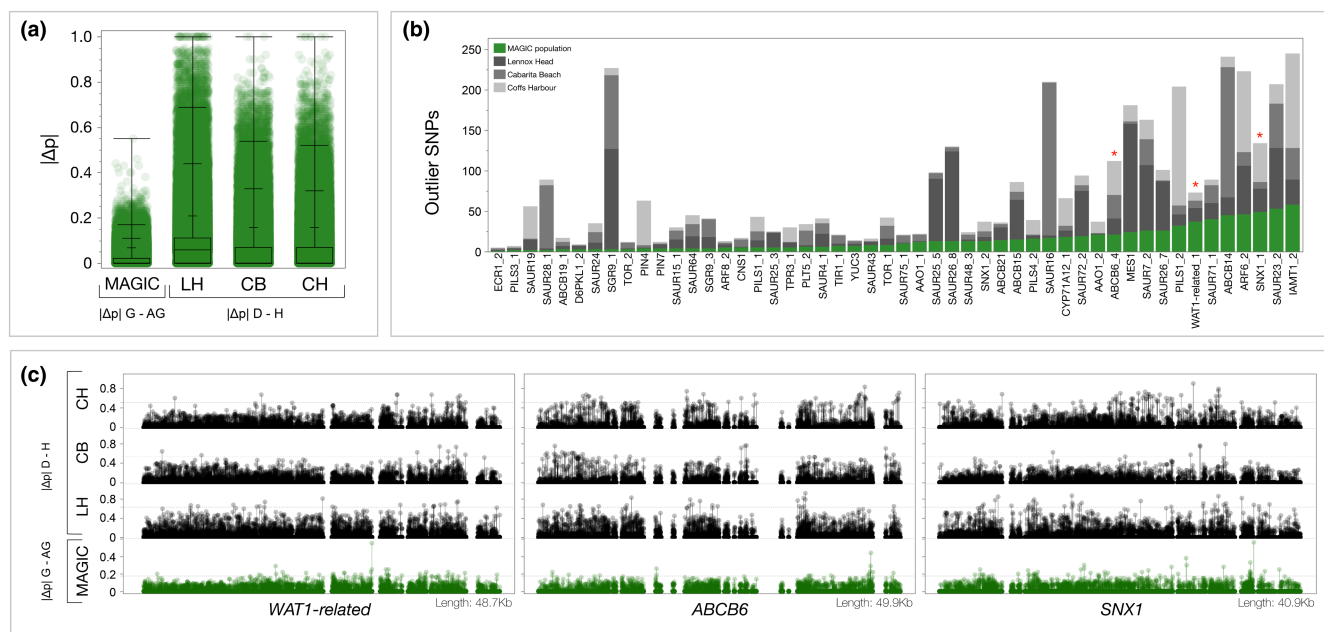


FIGURE 4 Reduction in genetic linkage in the MAGIC population. (a) The absolute change in allele frequencies ($|\Delta p|$) for variant sites of the 50 parallel architecture genes for the MAGIC (F11) population (gravitropic (G) and agravitropic (AG)) and the three natural population pairs (Dune (D) and Headland (H); Lennox Head—LH, Cabarita Beach—CB and Coffs Harbour—CH). The bars represent the 100, 99.5, 97.5, 90, 75 and 50 quantiles. (b) Number of outlier SNPs for each of the 50 parallel architecture genes for the MAGIC population and the three natural localities (also see Figure S10). The red asterisks correspond to the genes in (c). (c) Absolute allele frequency differences ($|\Delta p|$) in three natural localities (top three panels; Dune (D) and Headland (H); Lennox Head—LH, Cabarita Beach—CB and Coffs Harbour—CH) and the F11 MAGIC population (bottom panel; gravitropic (G) and agravitropic (AG)). Dashed grey lines represent the top 1% threshold of SNPs with the highest change in allele frequencies between the Dune and Headland (LH=0.63, CB=0.53, CH=0.52, MAGIC=0.18).

parallel evolution such as *S. laetus* enables us to overcome some of these limitations of nonmodel organisms to discover genomic components driving adaptive divergence.

Shared regions of elevated genomic divergence across multiple independent populations are unlikely to arise by random genetic drift (Losos, 2011; Orr, 2005; Schluter & Nagel, 1995). Here, we detected nine SNPs and 55 genes within a set of auxin genes that exhibited parallel divergence across all three *S. laetus* natural population pairs. These auxin SNPs and genes are strong candidates that have been repeatedly favoured by natural selection during the independent divergence of the Dune and Headland ecotypes. Remarkably, artificial tail selection in the MAGIC population largely reflected these patterns of replicated divergence—one of these nine parallel SNPs and 50 of these 55 parallel genes were outliers in the MAGIC population. These parallel architecture SNPs and genes are key candidates that link genetic variation in the auxin pathway to parallel adaptive trait divergence in *S. laetus*.

In natural populations, genetic divergence was often elevated for the entire length of outlier genes. For some genes, the MAGIC population enabled us to partially disassociate neutral variants and link narrow windows of genomic variation with gravitropism (e.g. *PIN4*, *SAUR28* and *SGR9*). This was facilitated by the large size of the MAGIC population: the base population consisted of 45 diploid individuals (F0), which enabled us to capture the genetic variation and allelic associations present in the natural populations. The repeated crossing for 11 generations (F11; $N_{\text{total}} = 1200$) further enabled us to break down genetic linkage. In genes where linkage did not break down in the MAGIC population (e.g. *SAUR23*), regions of low recombination caused by chromosomal inversions (Faria & Navarro, 2010) or recombination modifiers (Lenormand, 2012) might be responsible for these patterns. This could imply that genome structure might impose constraints on how natural selection favours adaptive alleles in nature (e.g. Samuk et al., 2017). Future and ongoing work in *S. laetus* to characterize genome structure and the genomic landscape of recombination will help elucidate the role of linkage in adaptive trait evolution.

In *S. laetus*, the genetic components underlying shoot gravitropism appear to be governed by a combination of parallelism (where some SNPs and genes have been repeatedly selected for) and redundancy (where replicate populations have adapted via somewhat unique genetic routes). In contrast to the very few parallel SNPs found in the auxin genes studied here, we observed striking patterns of replication in the auxin pathway at the level of the biological function. These biological functions give us insight into the physical mechanisms that enable the bending of a shoot during a gravitropic response. We found that the transport functional category (the active movement of auxin within and between cells) was consistently enriched in all natural localities as well as the MAGIC population. Previous work in *Arabidopsis* (Friml et al., 2003) and other plants (see Muday, 2001) suggests that gravitropic responses are facilitated by auxin transporters that create local auxin gradients in the stems and roots of plants. Our work thus further confirms the importance of auxin redistribution in eliciting gravitropic responses, and it validates

that the general machinery enabling a plant to bend might share similar molecular components across different plant species.

Genetic redundancy in *S. laetus* has implications for our ability to isolate signals of natural selection in the genes identified here. If genes are highly redundant in function, then selective constraints on a per-gene basis might be weak, and only in aggregate will they create signals of detectable polygenic adaptation (Barghi et al., 2020). Although our work shows that a range of genetic solutions underlie adaptive divergence across replicate populations, we place focus on outlier genomic regions that are common across replicate populations and the MAGIC population. These core regions have strong evidence of contributing to the repeated shoot gravitropism divergence in the *S. laetus* system, yet, represent a subset of all loci underlying gravitropism divergence. Rather than aiming to detect the entire genetic architecture of an adaptive trait, in this study we place an emphasis on how combining arguments for selection (e.g. parallel evolution) with genetic tools (e.g. a MAGIC population) is a powerful technique to isolate the core genetic components of parallel evolution for any key trait of interest. Our approach thus moves forward from more traditional studies of parallel evolution that are rarely able to directly link patterns of genomic divergence to the specific traits that underlie them.

The assembly and annotation of a chromosome-level genome is the focus of ongoing work in *S. laetus*. This will enable us to undertake detailed genome-wide association studies with the current dataset to elucidate the relative contributions of all genome-wide variants to divergent trait evolution, while estimating their effect size (Barghi et al., 2020; Sella & Barton, 2019). We can then estimate the percentage of phenotypic variation explained by parallel loci (Zeng et al., 2018) and further explore how genetic architectures might impose constraints on the direction of evolution by natural selection (Losos, 2011; Rosenblum et al., 2014; Yeaman et al., 2018). Although genetic knockouts and functional genomics will be required to confirm the functional role of the candidate genes we have identified, our work paves the way for future studies examining the relationship between genetic and fitness variation under natural conditions (e.g. Mitchell-Olds, 2013; Price et al., 2018; Schluter et al., 2021).

Overall, our research integrates replicate natural populations with the power of a MAGIC population to elucidate core genetic variation in a signalling pathway underlying the evolution of a parallel adaptive trait in nature. Our work provides insight into how deterministic processes contribute to the adaptation of organisms to similar selective pressures, how redundancy and functional equivalence can underlie complex genetic traits and how this impacts the levels of predictability in a system (Chevin et al., 2010; de Visser & Krug, 2014; Rosenblum et al., 2014; Yeaman et al., 2018). Our work brings us closer to understanding how parallelism arises at various levels of biological organization during the repeated evolution to common environmental challenges.

AUTHOR CONTRIBUTIONS

Daniel Ortiz-Barrientos and Melanie J. Wilkinson contributed to conceptualization; Maddie E. James, Daniel Ortiz-Barrientos,

Robin N. Allsopp, Avneet Kaur and Jeffrey S. Groh contributed to formal analysis; Robin N. Allsopp, Maddie E. James and Jeffrey S. Groh contributed to investigation; Maddie E. James and Melanie J. Wilkinson contributed to resources; Robin N. Allsopp, Maddie E. James, Melanie J. Wilkinson and Daniel Ortiz-Barrientos contributed to data curation; Maddie E. James and Daniel Ortiz-Barrientos contributed to writing—original draft and visualization; Maddie E. James, Daniel Ortiz-Barrientos, Robin N. Allsopp, Maddie E. James, Jeffrey S. Groh and Avneet Kaur contributed to writing—review and editing; Daniel Ortiz-Barrientos mentoring and supervision and funding acquisition.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Raw sequence data are available on UQ eSpace (<https://doi.org/10.48610/40dec7a>). Final data files and code are found on GitHub (<https://github.com/OB-lab/James-et-al-2023-MolecularEcology>).

ORCID

Maddie E. James  <https://orcid.org/0000-0001-6296-0187>

Robin N. Allsopp  <https://orcid.org/0000-0002-6802-4418>

Jeffrey S. Groh  <https://orcid.org/0000-0002-0444-4392>

Avneet Kaur  <https://orcid.org/0000-0001-5055-419X>

Melanie J. Wilkinson  <https://orcid.org/0000-0003-1400-3176>

Daniel Ortiz-Barrientos  <https://orcid.org/0000-0002-7493-416X>

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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